

RANJITH GOPI

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PROFESSIONAL SUMMARY

Generative AI Engineer with 3+ years of experience building cloud-native AI solutions, specializing in Retrieval-Augmented Generation (RAG) pipelines, vector databases, and LLM integrations using LangChain, OpenAI services, and semantic search. Skilled in developing scalable backend services and orchestrating cloud-based deployments and solutions with CI/CD and modern MLOps practices. Passionate about delivering secure, production-ready AI platforms that drive business impact through automation, performance optimization, and cost efficiency.

WORK EXPERIENCE

Generative AI Engineer, Cognizant Technology solutions Aug 2022 – Present

- Designed and developed scalable **RESTful APIs** with **Python + FastAPI** to power an enterprise document-aware Generative AI assistant, cutting **query resolution time by 30%** via semantically grounded responses.
- Built and deployed **Agentic RAG applications** using **LangChain, LangGraph, Azure OpenAI, and Azure AI Search**, improving **semantic retrieval accuracy by 30%** and enabling multi-agent orchestration for enterprise knowledge assistants.
- Engineered **multimodal Agentic RAG pipelines** (text, images, tables) to generate embedding-optimized summaries, reducing **storage and embedding costs by 40%** while ensuring compliance and auditability.
- Automated secure ingestion pipelines from **SharePoint Online → Azure Blob Storage** using Microsoft Graph API, Azure AD App Registrations, MSAL, with metadata-driven validation (hashing, deduplication, traceability) ensuring high-quality, governed data.
- Implemented **context-preserving text splitters** (hierarchical + recursive chunking) in LangChain, improving downstream **retrieval accuracy by 25%** for Agentic RAG workflows.
- Indexed and served 50K+ enterprise documents in **Azure AI Search (HNSW + semantic reranking)**, sustaining **sub-500ms query latency** at production scale.
- Applied **LLM grounding techniques** (retrieval reranking, structured prompting, guardrails) reducing **hallucinations by 22%** and boosting factual consistency in enterprise chat assistants.
- Enhanced **multi-turn reasoning** with persistent conversation memory in **CosmosDB**, leveraging **Zero-shot, few-shot prompting + Chain-of-Thought**, raising query resolution accuracy by **35%**.
- Productionized **GenAI microservices** with **Docker, Azure App Service, GitHub Actions CI/CD, and Azure Container Registry**, enabling **scalable, reliable deployments** of enterprise-grade SaaS AI solutions.
- Secured AI workloads using Azure Key Vault + managed identities, and implemented (Application Insights, structured logging), reducing **incident resolution time by 20%**.

SKILLS

Programming Languages & Frameworks

Python FastAPI, Pandas, NumPy, Streamlit, LangChain, LangGraph, Microsoft Graph API

Generative AI & LLMs

Azure OpenAI(GPT-4, GPT-4o), Prompt Engineering (Zero-shot, Few-shot, Chain-of-Thought), RAG, Agentic RAG (multi-agent orchestration),LLM Fine-Tuning

Vector Search & Embeddings

Azure AI Search (HNSW, Semantic Ranker), FAISS, Chroma DB, Azure OPenAI embeddings, Text Splitters, Vector Store Integration, Semantic Search, Cosine Similarity

Cloud Platforms & Azure Services

Azure Blob Storage, Azure Cosmos DB, Azure AI Search, Azure Key Vault, Azure App Service, Azure Container Registry (ACR), Docker, GitHub Actions, CI/CD Pipelines, Structured Logging

EDUCATION

Bachelor of Technology in Information Technology

2018 – 2022

Adhiyamaan College of Engineering, Hosur, TN

CGPA: 8.2 / 10

PROJECTS

McDonald's GenAI -RAG Project

June 2024 - Present *client Project*

- Developed scalable RESTful APIs with Python and FastAPI for a document-aware GenAI assistant, reducing query resolution time by 30% across enterprise-scale datasets.
- Implemented hierarchical and recursive text splitters in LangChain to handle complex document structures (PDFs, CSVs, tables), improving retrieval accuracy by 25% for downstream RAG workflows.
- Indexed and served 50K+ multi-format enterprise documents in Azure AI Search with HNSW + semantic reranking, sustaining sub-500ms query latency at production scale while maintaining compliance with data governance.
- Applied LLM grounding and structured prompting techniques, reducing hallucinations by 22% and improving factual consistency in enterprise chat assistants.
- Applied NLP preprocessing (normalization, intent classification, structured prompting), reducing hallucinations by 22%.
- Designed and integrated contextual caching and incremental embedding updates, reducing redundant computation and improving query throughput by 15% at peak load.

Gen AI Assurance Project

Jan 2023 - May 2024 *Internal Project*

- Built and deployed multi-agent RAG applications with LangChain and LangGraph, enabling dynamic multi-agent orchestration for enterprise knowledge assistants and boosting semantic retrieval accuracy by 30%.
- Engineered multimodal RAG pipelines for text, images, and tables, producing embedding-optimized summaries that reduced storage and embedding costs by 40% while ensuring full auditability.
- Designed and implemented advanced LLM grounding strategies to handle ambiguous queries, reducing hallucinations and improving factual consistency, enhancing user trust in AI-driven responses.
- Collaborated with cross-functional teams to integrate pipelines into enterprise workflows, ensuring scalable, maintainable, and compliance-ready deployments.
- Developed dynamic agent selection and orchestration logic, enabling the system to automatically route queries to the most relevant agent, improving response relevance by 20%.

DECLARATION

I hereby declare that all the information provided above is accurate to the best of my knowledge.

Ranjith Gopi